Ground-Based Remote Sensing Cloud Classification via Context Graph Attention Network

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***Abstract*— Most ground-based remote sensing cloud classification methods focus on learning representation features for cloud images while ignoring the correlations among cloud images. Recently, graph convolutional network (GCN) is applied to provide the correlations for ground-based remote sensing cloud classification, in which the graph convolutional layer aggregates information from the connected nodes of graph in a weighted way. However, the weights assigned by GCN cannot reflect the importance of connected nodes precisely, which declines the discrimination of the aggregated features (AFs). To overcome the limitation, in this article, we propose the context graph attention network (CGAT) for ground-based remote sensing cloud classification. Specifically, the context graph attention layer (CGA layer) of CGAT is proposed to learn the context attention coefficients (CACs) and obtain the AFs of nodes based on the CACs. We compute the CACs not only considering the two connected nodes but also their neighborhood nodes in order to stabilize the aggregation process. In addition, we propose to utilize two different transformation matrices to transform the node and its connected nodes into new feature spaces, which could enhance the discrimination of AFs. We concatenate the AFs with the deep features (DFs) as final representations for cloud classification. Since existing ground-based cloud data sets (GCDs) have limited cloud images, we release a new data set named GCD that is the largest one for ground-based cloud classification. We conduct a series of experiments on GCD, and the experimental results verify the effectiveness of CGAT.**

***Index Terms*— Context information, graph attention network (GAT), ground-based remote sensing cloud classification.**

I. Introduction

CLOUDS are the condensation of water vapor in the atmosphere, and they have a great influence on the hydrological cycle and energy balance on the earth [1]. The analysis of clouds can provide valuable information for human daily life, for example, weather prediction, climate prediction, and transportation [2], [3]. Consequently, a large number of cloud observation methods, such as satellite remote sensing observation [4]–[6], and ground-based remote sensing observation [2], [7], [8], are proposed for cloud-related research. The satellite remote sensing observation provides the overview of cloud systems and it directly observes the influence of clouds on the earth's radiation at the top of atmosphere. Therefore, it is appropriate for atmospheric research [9]. The target of ground-based remote sensing observation is to monitor the local area and characterize the bottom of clouds so as to obtain the detailed information of clouds [10]. This work mainly focuses on ground-based remote sensing observation, especially ground-based remote sensing cloud classification.

With the advent of ground-based remote sensing observation equipments, such as Whole Sky Imager (WSI) [11], Total Sky Imager (TSI) [12], and All Sky Imager (ASI) [8], a large number of cloud images are collected, which provides preconditions for the development of automatic cloud classification algorithms. Many ground-based remote sensing cloud classification methods [13]–[16] utilize handcrafted features to extract texture, color, and shape information from cloud images. However, the handcrafted features are predefined, and they cannot adapt to different data distributions.

Since convolutional neural network (CNN) could automatically learn discriminative features according to data distributions, it has become dominant in many research fields [17]–[19]. CNN also plays an important part in ground-based remote sensing cloud classification due to the promising performance [20]–[23]. Recently, some researchers [24] treat cloud images as the nodes of graph and apply the graph convolutional network (GCN) to learn the correlations for ground-based remote sensing cloud classification, in which the aggregated feature (AF) learned by GCN could aggregate the information from the cloud image itself and its connected images in a weighted way. However, the weight assigned by GCN is hard to reflect the importance of connected node precisely.

Graph attention network (GAT) [25] is a generalization of GCN, and it leverages the attention mechanism to pay more attention to the important connected nodes. GAT treats the attention coefficients as the weights and learns one transformation matrix to project the node and its connected nodes into a new feature space. Although GAT has the superiority of representation with the help of attention mechanism, there are two shortcomings when applying GAT to ground-based

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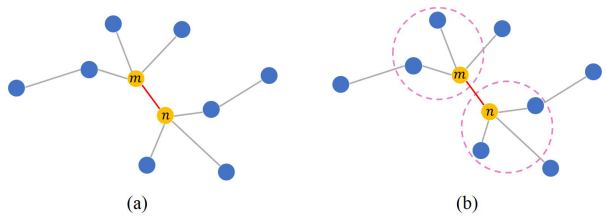


Fig. 1. Graphs in (a) GAT and (b) CGAT, where each node represents a cloud image. The red line indicates the attention coefficient between m and n, and the dashed circles represent the neighborhood nodes of m and n.

remote sensing cloud classification. First, since the cloud image is a kind of nature texture, which is easily disturbed by environmental factors such as illumination and noise, the attention coefficient computed based on only two nodes is unstable for the aggregation process, as shown in Fig. 1(a). Second, the node and its connected nodes are different, but they share the same transformation matrix, which results in decreasing the discrimination of the AFs.

To overcome the above-mentioned drawbacks, in this article, we propose a novel deep network termed context graph attention network (CGAT) for ground-based remote sensing cloud classification. Specifically, the proposed CGAT first utilizes CNN to extract the deep features (DFs) of cloud images and then constructs a graph based on these DFs. Afterward, the graph is fed into the context graph attention layer (CGA layer) to learn the AFs for the nodes. In the aggregation process, we propose the context attention coefficient (CAC) and serve it as the weight to enhance the stability of the AFs. The CAC not only considers the two nodes themselves but also the context information in their neighborhood nodes. As shown in Fig. 1(b), both nodes m and n and their neighborhood nodes, which are selected from their connected nodes in the graph, are involved to compute the CAC between nodes m and n. Moreover, we propose to learn two different transformation matrices to transform the node and its connected nodes, respectively, which could improve the discrimination of the AFs.

In addition, a large data set is necessary for the development of ground-based remote sensing cloud classification algorithms. Especially for the deep neural network, a large data set can avoid overfitting and improve the model generalization. However, the existing public ground-based cloud image data sets only contain thousands of cloud images, which may restrict the development of cloud classification algorithms. For example, Singapore Whole-sky IMaging CATegories (SWIMCAT) [27], Kiel [14], Cirrus Cumulus Stratus Nimbus (CCSN) [21], and Total-sky Cloud Imager Dataset (TCID) [26] contain 784, 1500, 2543, and 5000 cloud images, respectively. In this article, we release the Ground-based Cloud data set (GCD) that contains 19 000 cloud images. To the best of our knowledge, GCD is the largest GCD and it is almost four times larger than the existing data sets. Furthermore, it is collected in ten provinces of China over one year, so it possesses large diversity.

There are three contributions of the proposed CGAT. First, we propose the CAC for the aggregation process, which not only relies on two nodes themselves but also their neighborhood nodes. As a result, the stability of the AFs is enhanced. Second, we propose to apply two transformation matrices to transform the node and its connected nodes in order to obtain the discriminative AFs. Finally, we collect a new data set GCD, which contains 19 000 cloud images, and the proposed CGAT outperforms state-of-the-art methods on GCD, which verifies the effectiveness of CGAT.

II. Related Work

A. Traditional Methods for Ground-Based Remote Sensing Cloud Classification

Traditional methods for ground-based remote sensing cloud classification are usually based on handcrafted features of cloud images. They are roughly divided into two categories. The first category methods directly employ handcrafted features to represent cloud images. For example, Zhuo *et al.* [28] proposed the color census transform (CCT) to preprocess cloud images and then computed a histogram and image statistics to obtain feature vectors for cloud representation. Oikonomou *et al.* [16] employed the regional local binary pattern (R-LBP) and four patch-local binary pattern (FP-LBP) to extract global and local texture features to represent cloud images.

The second category methods apply handcrafted features to generate textons and then obtain cloud representations based on the textons. For example, Liu *et al.* [29] presented multiple random projections (MRPs) to discover different parts of cloud image structure so as to obtain the optimal textons and then extracted cloud features using these textons. Dev *et al.* [27] proposed to generate the modified textons using the s-filters that are consisted of several rotation-invariant Gabor-like filters.

B. Deep Learning Methods for Ground-Based Remote Sensing Cloud Classification

CNN has shown its superiority in many research fields, and it is extended to ground-based remote sensing cloud classification [30]–[32]. Some methods utilize CNN to classify cloud images in an end-to-end way, which takes cloud images as input and the classification results as output. The 3-D-CNN model [23] was proposed to extract textural and temporal information of cloud images and utilized a fully connected (FC) layer to output the classification results. Zhao *et al.* [32] presented the multichannel CNN to extract cloud features from multiple channels and then export the classification results of cloud images.

Meanwhile, some cloud classification methods apply CNN as the extractor that is they extract cloud features from CNN and then feed the cloud features into the independent classifier. Shi *et al.* [30] applied the outputs of shallow convolutional layer of CNN as cloud features and then employed support vector machine (SVM) to decide the cloud type. Ye *et al.* [31] extracted the multiscale feature maps from the pretrained CNN and then employed the Fisher vector (FV) to encode them. Finally, SVM is treated as the classifier.

C. GCN and GAT

GCN is a successful generalization of CNN from Euclidean data to graph-structured data [33]. Since various types of data

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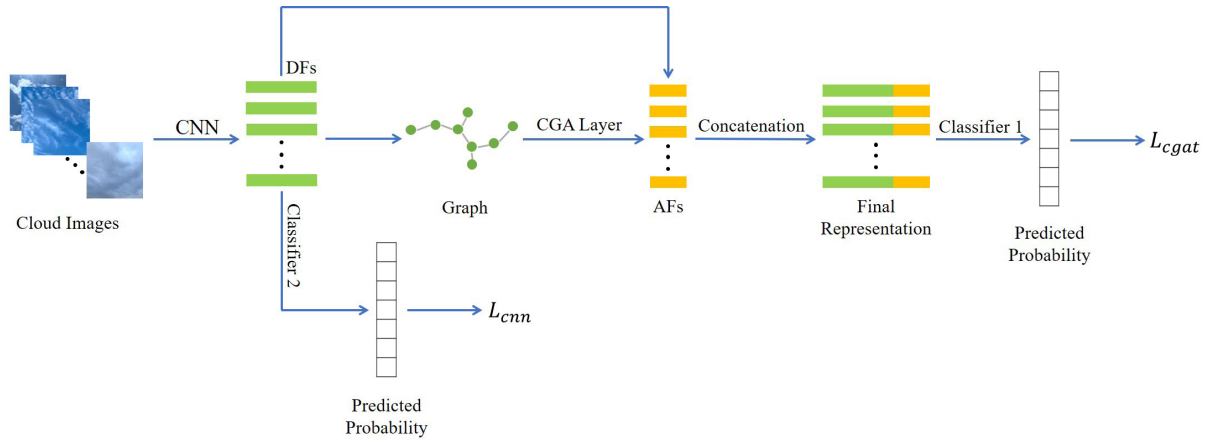


Fig. 2. Framework of the proposed CGAT. ResNet-50 is treated as CNN in our framework. Classifier 1 includes two FC layers with 1024 and T neurons, respectively, and Classifier 2 includes one FC layer with T neurons, where T is the number of cloud types. Both L_{cgat} and L_{cnn} are the cross-entropy losses.

can be changed to the graph form, GCN has been applied in many research fields, such as text classification, 3-D human pose regression, and road traffic speeds forecast [34]–[36]. GCN has obtained promising performance in these fields, and it is also applied to ground-based remote sensing cloud classification. For example, Liu *et al.* [24] proposed a task-based graph convolutional network (TGCN) for ground-based remote sensing cloud classification, in which the graph is built based on DFs and DFs that are supervised by the classification task. However, the weights assigned by GCN are hard to reflect the importance of connected node in the graph precisely.

To overcome the above-mentioned shortcoming, GAT is proposed. It is constructed based on the attention mechanism and aggregates the information from the connected nodes using the attention coefficients [25]. Here, the attention coefficients are treated as the weights in the aggregation process. As a result, GAT could obtain a precise graph representation and it plays an important role in many research fields [37]–[39]. For example, Kosaraju *et al.* [37] formulated pedestrian interactions as a graph for multimodal trajectory forecasting, where each pedestrian in the graph interacts with others using different attention coefficients. Li *et al.* [39] learned the intrarelation and inter-relation between action proposals according to the graph attention mechanism for the action detection task. In order to learn stable and discriminative cloud features, in this work, we propose a kind of GAT named CGAT for ground-based remote sensing cloud classification, which learns CACs and different transformation matrices.

III. Approach

A. Overview of CGAT

In this section, we give an overview of CGAT and the framework of CGAT is shown in Fig. 2. The proposed CGAT is composed of graph construction, CGA layer, and final representation and classification.

1) Graph Construction: We first learn the DFs of cloud images and then compute the similarities of these DFs. We treat the DFs of cloud images as the nodes of graph and link the similarities larger than a predefined threshold as the edges of graph.

2) Context Graph Attention Layer: After constructing the graph, we propose the CGA layer to extract the AFs. The CGA layer first computes the CAC to provide the context information of the node and its neighborhood nodes that are selected from the connected nodes in the graph. Then, the CAC is applied to aggregate the information from the connected nodes. Furthermore, the CGA layer learns two transformation matrices to project the nodes and their connected nodes to different feature spaces.

3) Final Representation and Classification: The AFs extracted by the CGA layer contain rich correlations among cloud images, and meanwhile, the DFs capture the information of cloud structure and texture. In order to represent the cloud image completely, we concatenate the AF and the DF as the final representation for cloud classification. In CGAT, two classifiers are applied to obtain the predicted probabilities of cloud images, and two loss functions are utilized to train the overall framework of CGAT.

B. Graph Construction

The graph $G(V, E)$ is consisted of a set of nodes $V = \{v_1, v_2, \ldots, v_N\}$ and a set of edges $E = \{e_{\text{ij}}\}$, where N is the number of nodes in the graph, and the node $v_i \in \mathbb{R}^d$ is a feature vector with the dimension of d.

In this work, we utilize the cloud images to construct the graph. Specifically, we extract the DFs of cloud images from CNN and treat them as the nodes of graph. Here, ResNet-50 [40] is employed as CNN and its framework is shown in Fig. 3, in which the last FC layer is eliminated. $Conv2_x$–$Conv5_x$ contain 3, 4, 6, and 3 building blocks, respectively, and each building block includes three convolutional layers. For example, each building block in $Conv5_x$ contains three convolutional layers with the kernel size of 1×1, 3×3, and 1×1, and the number of feature maps are 512, 512, and 2048, respectively. We resize the cloud image into 256×256 and feed them into ResNet-50. Afterward, we obtain 2048 feature maps with the size of 8×8 from the final

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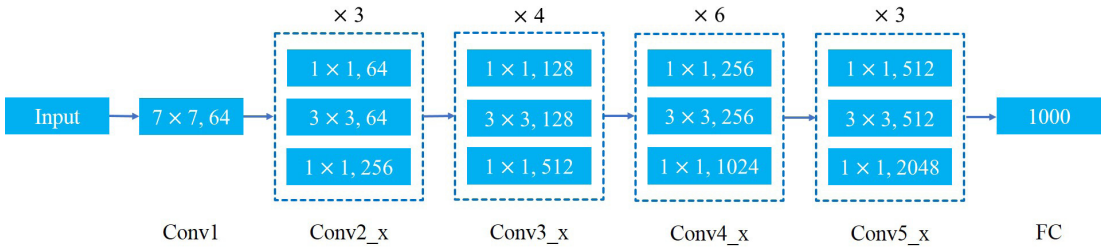


Fig. 3. Structure of ResNet-50. *Conv*2_*x*–*Conv*5_*x* are four different building blocks in ResNet-50. The numbers in the boxes of *Conv*1 and *Conv*2_*x*–*Conv*5_*x* are the kernel size and the number of feature maps of convolutional layers, and the last FC layer contains 1000 neurons. The numbers above the boxes are the number of building blocks.

convolutional layer in *Conv*5_3 of ResNet-50 and perform the average pooling to obtain the DFs with the dimension of 2048, i.e., $d = 2048$.

In order to link the nodes in the graph, we compute the similarity between two nodes

$$s_{\mathrm{ij}} = {v_i}^T v_j \tag{1}$$

where v_i and v_j are node i and node j, respectively. Then, we apply the predefined threshold τ on these similarities to obtain the edge between nodes i and j

$$e_{\mathrm{ij}} = \begin{cases} 1, & s_{\mathrm{ij}} \geq \tau \\ 0, & s_{\mathrm{ij}} < \tau. \end{cases} \tag{2}$$

We utilize the adjacency matrix A to represent the correlations in the graph, and A is defined as

$$A = \begin{pmatrix} e_{11}, & e_{12}, & \cdots, & e_{1N} \\ e_{21}, & e_{22}, & \cdots, & e_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ e_{N1}, & e_{N2}, & \cdots, & e_{NN} \end{pmatrix} \tag{3}$$

where the edge e_{ij} is equal to 1 or 0 indicating whether nodes i and j are connected or not. Based on 2, we can suppress the redundant edges that have low similarities between the nodes. As a result, the graph is locally connected and it could avoid the noise interference from those redundant edges.

C. Context Graph Attention Layer

After constructing the graph, GCN and its variants [24], [33] can be employed to learn correlations among cloud images so as to obtain valuable information from other cloud images. In this section, we first review GAT [25] which is a representative variant of GCN, and then, we introduce the proposed CGA layer, including the CAC and the AF.

1) Review of GAT: GAT applies the attention mechanism to learn the attention coefficient a_{ij} between nodes i and $j \in P_i$, where P_i indicates a set of nodes connected to node i in the graph and it also includes i. Then, a_{ij} is served as the weight to aggregate the information from the connected nodes. Specifically, the attention coefficient a_{ij} is defined as

$$a_{\mathrm{ij}} = \varepsilon(t(v_i, v_j)) = \frac{e^{t(v_i, v_j)}}{\sum_{k \in P_i} e^{t(v_i, v_k)}} \tag{4}$$

where

$$t(v_i, v_j) = (U v_i) \cdot (V v_j)^T \tag{5}$$

where $t(,)$ indicates the attention mechanism, $\varepsilon(\cdot)$ is the softmax function that is used to normalize the attention coefficients, and $U, V \in \mathbb{R}^{d_u \times d}$ are the learnable weight matrices. The AF of node i in GAT is formulated as

$$g_i = h\left(\sum_{j \in P_i} a_{\mathrm{ij}} v_j W\right) \tag{6}$$

where $W \in \mathbb{R}^{d \times d_w}$ is the transformation matrix to project the nodes in the graph to a new feature space and $h(\cdot)$ indicates the activation function implemented by rectified linear unit (ReLU).

2) Context Attention Coefficient: The attention coefficient learned by GAT reflects different importance of each connected node, which is of great importance for aggregating related information. However, since the cloud image is a kind of nature texture, which is easily disturbed by environmental factors, the attention coefficient computed on two nodes is unstable for the aggregation process. Furthermore, the transformation matrix transforms all the nodes in the graph without distinguishing the nodes and their connected nodes, which decreases the discrimination of AFs. To overcome the limitations, we propose the CGA layer, which obtains the CAC based on the neighborhood nodes and aggregates the features using different transformation matrices.

The CAC is served as the weight to aggregate the information from the connected nodes. The CAC not only relies on two nodes but also on their neighborhood nodes that are selected from their connected nodes in the graph. In this way, the context information ensures the precision of the CAC, which is beneficial to obtain the stable AFs. The CAC is defined as

$$c_{\mathrm{ij}} = \xi(\sigma(v_i, v_j)) = \frac{e^{(\sigma(v_i, v_j) - m)}}{\sum_{k \in P_i} e^{(\sigma(v_i, v_k) - m)}} \tag{7}$$

where

$$m = \max_k (\sigma(v_i, v_k)) \tag{8}$$

where $\xi(\cdot)$ indicates the modified softmax function, which can avoid the problem of numerical instability, and m indicates the maximum value of $\sigma(v_i, v_k)$, which could avoid quite large or

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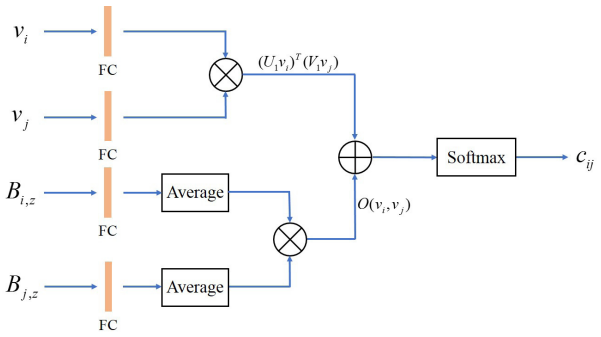


Fig. 4. Diagram of CAC c_{ij} between nodes i and j. $B_{i,z}$ and $B_{j,z}$ are the selected neighborhood nodes of nodes i and j, and "$\oplus$" and "$\otimes$" represent the operations of addition and multiplication, respectively. FC indicates the FC layer, and the four FC layers from top to bottom correspond to U_1, V_1, U_2, and V_2, respectively.

small value of the exponential term. Here, $\sigma(v_i, v_j)$ is the context attention mechanism and it is formulated as

$$\sigma(v_i, v_j) = (U_1 v_i)^T (V_1 v_j) + O(v_i, v_j) \tag{9}$$

where $U_1 \in \mathbb{R}^{d' \times d}$ and $V_1 \in \mathbb{R}^{d' \times d}$ are the learnable weight matrices that are realized by the FC layers. $O(v_i, v_j)$ considers the context information of the two nodes v_i and v_j

$$O(v_i, v_j) = \frac{(U_2 \sum_{b \in B_{i,z}} v_b)^T (V_2 \sum_{r \in B_{j,z}} v_r)}{z^2} \tag{10}$$

where $U_2 \in \mathbb{R}^{d' \times d}$ and $V_2 \in \mathbb{R}^{d' \times d}$ are the learnable weight matrices realized by the FC layers, and $B_{i,z}$ and $B_{j,z}$ are the sets of z neighborhood nodes of nodes i and j, respectively. Here, the neighborhood nodes are selected based on the similarity from the connected nodes in the graph. Fig. 4 shows the process of 7.

3) Aggregated Features: After obtaining the CACs, we learn the AFs based on the connected nodes and the CACs. Specifically, the AF of node i is defined as

$$g_i = h(v_i W_1 + \sum_{j \in P_i, j \neq i} c_{ij} v_j W_2) \tag{11}$$

where $W_1, W_2 \in \mathbb{R}^{d \times d_w'}$ are the transformation matrices. Here, we utilize two transformation matrices to project the node itself and its connected nodes into different feature spaces in order to improve the discrimination of AFs.

In addition, we employ the multihead attention strategy in the CGA layer to obtain the completed AFs. The multihead attention strategy learns multiple CACs independently, and then, these CACs are applied to obtain the AFs. These AFs are concatenated to obtain the multihead attention-based AFs

$$g_i = \|_{y=1}^{Y} h(v_i W_1 + \sum_{j \in P_i, j \neq i} (c_{ij})_y v_j W_2) \tag{12}$$

where $\|$ indicates the operation of concatenation, Y is the number of the independent CACs, and $(c_{ij})_y$ is the yth CAC between nodes i and j.

D. Final Representation and Classification

The AF obtained by CGAT possesses rich correlations among different cloud images, whereas the DF learned by CNN contains structure and texture information of individual cloud image. Consequently, we concatenate the DF and the AF as the final representation to obtain accurate descriptions of the cloud images. Specifically, the final representation is defined as

$$f_i = (g_i)^T \| \mu {v_i}^T \tag{13}$$

where μ is the parameter to balance the importance of DF.

As shown in Fig. 2, we apply two classifiers in CGAT. Classifier 1 consists of two FC layers that have 1024 and T neurons, respectively. Here, T is the number of cloud types. The first FC layer is followed by a Leaky ReLU activation function, and the last FC layer is followed by a modified softmax function. Classifier 1 outputs the predicted probabilities of the cloud image belonging to each cloud type. Afterward, the cross-entropy loss followed by Classifier 1 is applied to optimize CGAT:

$$L_{\text{cgat}} = -\sum_{t=1}^{\Phi} q_t \log y_t \tag{14}$$

where Φ indicates the neuron number of Classifier 1, q_t is the real probability of the cloud image belonging to the tth cloud type, and $q_t = 1$ when t is the ground-truth label; otherwise, $q_t = 0$. Here, $y_t \in [0, 1]$ is the output of the modified softmax function in Classifier 1, and it indicates the predicted probability of the cloud image belonging to the tth cloud type. Meanwhile, we also employ the cross-entropy loss L_{cnn} combined with Classifier 2 to optimize CNN. Classifier 2 contains an FC layer followed by a modified softmax function, and the FC layer has Φ neurons. As a result, the objective function of CGAT is formulated as

$$L = L_{\text{cgat}} + \lambda L_{\text{cnn}} \tag{15}$$

where λ is the parameter to balance the importance of L_{cnn} in the objective function.

IV. Experiments

In this section, we carry out a series of experiments to evaluate the performance of CGAT. We first introduce the released data set, i.e., GCD, and then describe the implementation details and the evaluation criterion in our experiments. Afterward, to verify the effectiveness of CGAT, we show the evaluation performance, including the ablation study of CGAT and comparison with other methods. Finally, several important parameters are analyzed.

A. Data Set

Since the existing public GCDs are insufficient for training deep neural models, we collect a new data set GCD, including 19 000 cloud images, which is the largest GCD. The cloud images are captured by camera sensors in nine provinces of China, i.e., Tianjin, Anhui, Sichuan, Gansu, Shandong, Hebei, Liaoning, Jiangsu, and Hainan. GCD is collected

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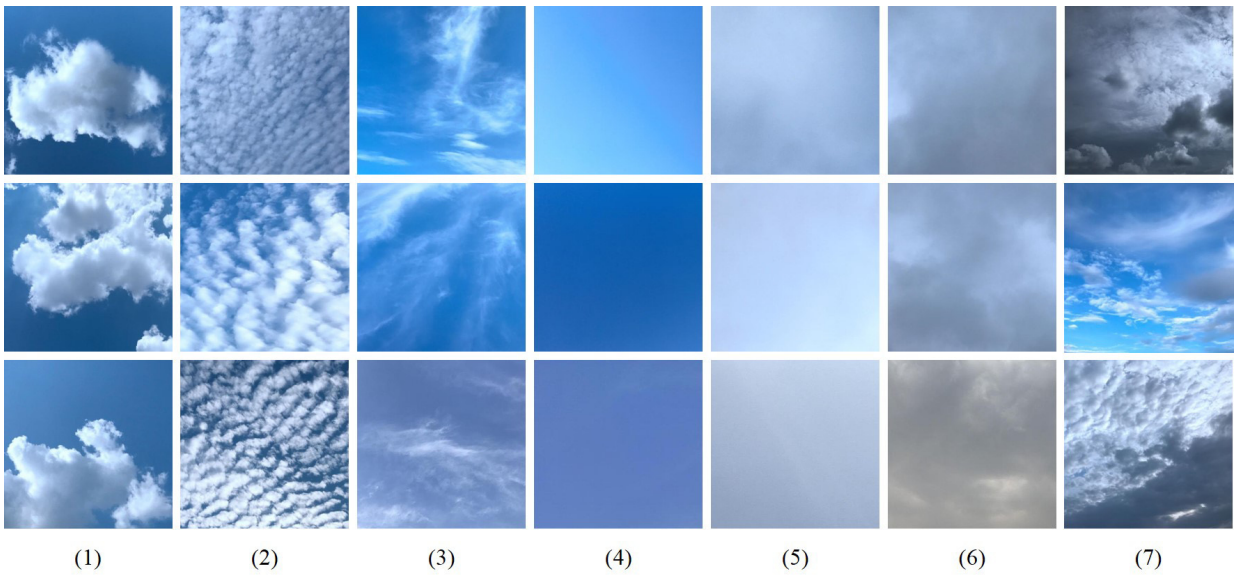


Fig. 5. Cloud images with seven types in GCD. (1) cumulus, (2) altocumulus and cirrocumulus, (3) cirrus and cirrostratus, (4) clear sky, (5) stratocumulus, stratus and altostratus, (6) cumulonimbus and nimbostratus, and (7) mixed cloud.

TABLE I
CHARACTERISTICS OF EACH CLOUD TYPE IN GCD

Cloud Type	Descriptions
Cumulus	Puffy clouds with clearly defined edges, flat bottom and raised top, white or light-grey
Altocumulus and Cirrocumulus	Patched clouds of small cloudlets, mosaic-like, white or grey
Cirrus and Cirrostratus	Thin clouds, fibrous, white feathery clouds of ice crystals or sky covering, whitish
Clear Sky	No cloudiness or a very few cloudiness, blue
Stratocumulus, Stratus and Altostratus	Lumpy layer of clouds, broken to almost overcast, causes fog or fine precipitation, white or grey
Cumulonimbus and Nimbostratus	Dark, thick clouds, mostly overcast, indistinct outline, vigorous vertical development, thundercloud or rain cloud, grey
Mixed	More than one type of clouds

TABLE II
NUMBER OF CLOUD IMAGES FOR EACH TYPE IN GCD

Cloud Type	Label	Sample Number
Cumulus	1	1525
Altocumulus and Cirrocumulus	2	1475
Cirrus and Cirrostratus	3	1906
Clear Sky	4	3739
Stratocumulus, Stratus and Altostratus	5	3636
Cumulonimbus and Nimbostratus	6	5764
Mixed	7	955
Total		19000

over one year, so it possesses large diversity. GCD is available at https://github.com/shuangliutjnu/TJNU-Ground-based-Cloud-Dataset. GCD contains seven cloud types that are divided according to the classification criterion released by the World Meteorological Organization (WMO) [41]. Fig. 5 shows some cloud images with seven types, including (1) cumulus, (2) altocumulus and cirrocumulus, (3) cirrus and cirrostratus, (4) clear sky, (5) stratocumulus, stratus, and altostratus, (6) cumulonimbus and nimbostratus, and (7) mixed cloud. Table I lists the characteristics of seven cloud types. The cloud images in GCD are with the resolution of 512×512 and they are stored in the Joint Photographic Experts Group (JPEG) format. GCD is divided into 10 000 training cloud images and 9 000 test cloud images. In order to make sure that there is no overlap between the training set and the test set, we select 10 000 cloud images from the first six months as the training set and select 9 000 cloud images from the second six months as the test set. We list the number of cloud images for each type in Table II.

B. Implementation Details and Evaluation Criterion

We resize the cloud images into 256×256 and conduct the random horizontal flipping with the probability of 0.5. In addition, we normalize the cloud images by subtracting mean values and dividing the standard deviation values, where the mean values and standard deviation values of RGB channels are computed based on all training cloud images in GCD.

The CNN model in CGAT is initialized by the pretrained ResNet-50, and the weights of all FC layers in CGAT are initialized by a standard normal distribution and the biases are initialized by 0. We apply two CGA layers and employ the multihead strategy in the last CGA layer, in which two independent CACs are learned. The threshold τ utilized to construct the graph in 2 is set to 0.7, and the number of

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TABLE III
COMPARISONS WITH DIFFERENT ABLATION METHODS, WHERE "w/o" INDICATES "WITHOUT"

Method	Accuracy (%)
w/o Graph	74.33
w/o Locally-Connected	82.94
w/o CAC	81.75
w/o W_1/W_2	81.98
w/o Multi-head	82.36
w/o Concatenation	82.65
CGAT	**84.30**

TABLE IV
COMPARISONS WITH OTHER METHODS

Method	Accuracy (%)	Method	Accuracy (%)
BoW	61.88	CloudNet [21]	74.84
PBoW	61.23	ResNet-50 [40]	74.33
$\text{LBP}_{(8,1)}$	45.68	VGG-19 [42]	75.11
$\text{LBP}_{(16,2)}$	51.42	DeepCloud [31]	74.70
$\text{LBP}_{(24,3)}$	52.30	DGL [43]	75.33
$\text{CLBP}_{(8,1)}$	54.73	TGCN [24]	81.03
$\text{CLBP}_{(16,2)}$	56.84	CGAT_VGG-19	83.60
$\text{CLBP}_{(24,3)}$	52.01	CGAT	**84.30**

neighborhood nodes for providing the context information z in 10 is set to 3. In the training stage, the number of epochs is set to 35, and the batch size is 32. The initial learning rate is set to 3×10^{-4} and it is decayed at epochs 15 and 25 with the decay rate of 0.25.

In our work, we employ the classification accuracy as the evaluation criterion to compare with other methods. The classification accuracy is defined as

$$\text{Accuracy} = \frac{\sum_{t=1}^{T} r_t}{Q} \tag{16}$$

where r_t is the number of correctly classified cloud images in the tth cloud type and Q indicates the total number of cloud images in the test set.

C. Performance Evaluation

1) Ablation Study of CGAT: The strength of CGAT lies in learning stable and discriminative AFs to describe cloud images accurately. The strength is guaranteed by several components of CGAT, i.e., locally connected graph, CAC, two different transformation matrices, multihead strategy, and concatenation DFs and AFs. In order to show their contributions on the performance improvement of CGAT, we conduct several experiments on the ablation methods of CGAT, and the classification results are shown in Table III.

From Table III, several conclusions can be drawn. First, the proposed CGAT obtains the best classification accuracy, which verifies the effectiveness of our method. Second, all the methods significantly surpass "w/o Graph," which only applies DFs as the final representations. It is because the correlations among cloud images are important to enhance the discrimination of cloud representation. Third, CGAT exceeds "w/o CAC," "w/o W_1/W_2," and "w/o Multihead" by 2.55%, 2.32%, and 1.94%, respectively. It demonstrates that the CAC, the two different transformation matrices, and the multihead strategy are the key components of CGAT to capture precise correlations among cloud images. Fourth, CGAT achieves higher classification accuracy than "w/o Locally Connected" because the locally connected graph could eliminate the noise from the redundant nodes. Finally, "w/o Concatenation" only employs the AFs extracted from CGAT as the final representation of cloud image, and its performance is declined. It is because the concatenation of AFs and DFs contains rich information, including the correlations among cloud images, texture, and structure information of cloud images.

2) Comparison With Other Methods: In this section, we compare CGAT with other methods and the classification accuracies are listed in Table IV. These compared methods contain traditional methods and CNN-based methods. The traditional methods extract handcrafted features from cloud images, including bag-of-words (BoW) [44], pyramid BoW (PBoW) [45], local binary patterns (LBPs) [46], and completed LBP (CLBP) [47]. BoW first extracts scale invariant feature transform (SIFT) features of cloud images and then employs the K-means algorithm on these SIFT features to generate codewords. Finally, the cloud feature is obtained by the frequency statistic of SIFT features on the codewords, and the number of codewords in our experiment is set to 300. PBoW provides spatial information for cloud images, and it divides each cloud image into 21 cells with three scales. The frequency statics based on the codewords are computed on each cell, and then, these frequency statics are concatenated as the final cloud feature with the dimension of 6300.

LBP generates cloud features according to the LBP values that are calculated based on gray value differences between central pixel and neighborhood pixels in local regions of a cloud image. CLBP is an improvement of LBP, and it considers the sign and magnitude of gray value difference, and the binary value of central pixels. LBP and CLBP both apply the rotation-invariant uniform patterns. As for LBP and CLBP, we set the number of neighborhood pixels P and radius of local region R to (8, 1), (16, 2), and (24, 3). Besides these traditional methods, we also apply the CNN-based methods for comparison, for example, CloudNet, ResNet-50, Visual Geometry Group (VGG)-19, dual guided loss (DGL), and TGCN, and these methods are the state-of-the-art methods.

From Table IV, several conclusions can be summarized as follows. First, the proposed CGAT surpasses other methods. Second, the CNN-based methods in the right of Table IV perform better than traditional methods in the left because the CNN model could learn highly nonlinear transformation according to different data distributions so that the cloud

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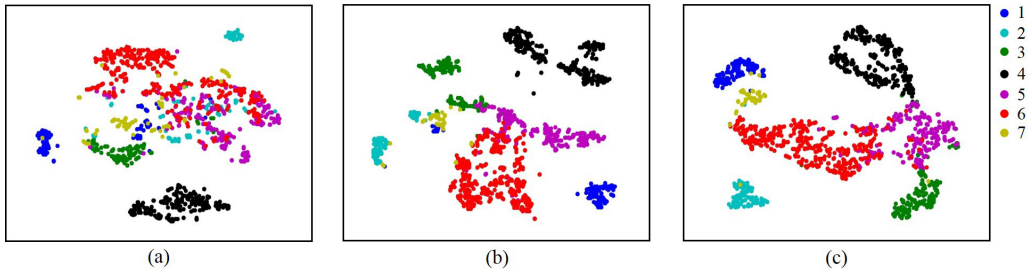


Fig. 6. Feature visualizations using t-distributed stochastic neighbor embedding (t-SNE) [48], in which each dot represents the final feature of the cloud image, and different colors indicate different cloud types. Feature visualizations of (a) CloudNet, (b) TGCN, and (c) CGAT.

features are discriminative. Third, TGCN and CGAT both have more significant improvement than other state-of-the-art methods. It is because TGCN and CGAT provide the correlations among cloud images that are of great importance for representing cloud images. Fourth, CGAT performs better than TGCN because the proposed context attention layer considers the context information and two transformation matrices are utilized to enhance the discrimination of AFs. Finally, CGAT_VGG-19 utilizes VGG-19 to replace RseNet-50 in CGAT, and CGAT obtains higher classification result than CGAT_VGG-19. It is because the deeper neural network ResNet-50 is more suitable as the backbone of CGAT than VGG-19.

In order to verify the effectiveness of CGAT intuitively, we visualize the final features of different methods as shown in Fig. 6, where Fig. 6(a)–(c) are the feature visualizations of CloudNet, TGCN, and CGAT, respectively. We can see that the dot with the same color is more closely in Fig. 6(c) because the final features of CGAT not only contain structure and texture information of the cloud image itself but also the context information of other cloud images. Hence, CGAT can describe cloud images accurately.

Furthermore, we also compute the confusion matrix to show the classification performance of each cloud type for CGAT in Fig. 7, where Fig. 7(a) shows the prediction results of CGAT and Fig. 7(b) shows the ground truth. For example, in Fig. 7(a), the nonzero values in the first row indicate that 91% cloud images of the first type are accurately classified, and 4% and 5% cloud images of the first type are misjudged as the second cloud type and the seventh cloud type, respectively. We can see that the fourth cloud type achieves the highest classification accuracy, whereas the seventh cloud type has the lowest accuracy. It is because the fourth cloud type is the clear sky that possesses low diversity, and the seventh cloud type is the mixed cloud type that is relatively difficult to distinguish.

In Fig. 7(a), we can see that some off-diagonal elements are nonzero, and they are the interclass misclassification in our method. The interclass misclassification is mainly caused by two reasons. First, the prediction of cloud types might be influenced by the variations in illumination and viewpoint. In Fig. 8, we select two misjudged cloud images Fig. 8(a) and (c) that both belong to the first type, but they are misjudged as the sixth type and the second type, respectively. Specifically, Fig. 8(a) is misjudged as the sixth type because the illumination causes the bottom of cloud to change to dark

	1	2	3	4	5	6	7
1	0.91	0.04	0.00	0.00	0.00	0.00	0.05
2	0.00	0.85	0.00	0.00	0.03	0.00	0.12
3	0.00	0.00	0.87	0.00	0.05	0.00	0.08
4	0.00	0.00	0.00	1.00	0.00	0.00	0.00
5	0.00	0.18	0.00	0.00	0.77	0.05	0.00
6	0.00	0.00	0.00	0.00	0.15	0.80	0.05
7	0.04	0.21	0.04	0.00	0.00	0.01	0.70

(a)

	1	2	3	4	5	6	7
1	1.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0.00	1.00	0.00	0.00	0.00	0.00	0.00
3	0.00	0.00	1.00	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	1.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	1.00	0.00	0.00
6	0.00	0.00	0.00	0.00	0.00	1.00	0.00
7	0.00	0.00	0.00	0.00	0.00	0.00	1.00

(b)

Fig. 7. (a) Confusion matrix of CGAT, which shows the predicted results on the test set. (b) Ground truth of the test set.

black and it is similar to the sixth type in Fig. 8(b). Fig. 8(c) is misjudged as the second type because clouds are moving and it causes the change of shooting viewpoint. Hence, the captured cloud in Fig. 8(c) looks like the second cloud type in Fig. 8(d). Second, the GCD possesses high diversity because it is the largest GCD and it is collected in ten provinces of China over one year. As a result, it is not easy to obtain a relatively high classification accuracy.

3) Parameter Analysis: In this section, we evaluate the influences of several important parameters for the proposed CGAT. These parameters include the number of CGA layers, the threshold τ in 2, the number of neighborhood nodes z, the number of CAC, i.e., Y, in the multihead strategy, the coefficient μ in 13, and the coefficient λ in 15.

a) Layer Number of CGA Layer: In this work, we propose the CGA layer to learn the AFs of cloud images. We conduct experiments with a different number of CGA layers to show its impact on the classification accuracy.

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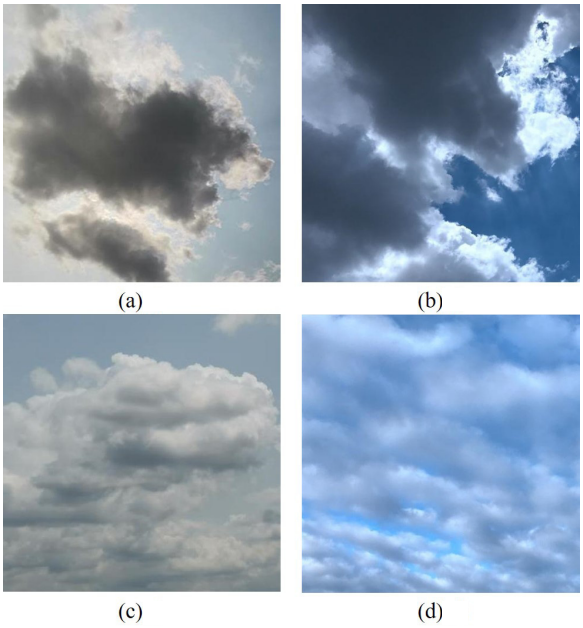


Fig. 8. Some cloud images from the test set. (a) and (c) First type of clouds. (b) and (d) Sixth and second cloud type.

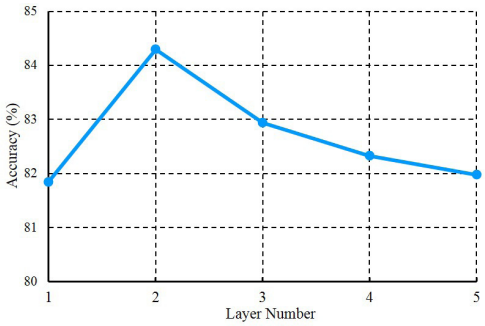


Fig. 9. Classification accuracies of CGAT with a different number of CGA layers.

Fig. 9 shows the variation of classification accuracies with a different number of CGA layers, and it can be concluded that the proposed CGAT obtains the highest classification accuracy when the layer number is set to 2.

b) Threshold τ for Constructing the Locally Connected Graph: In our work, we utilize the locally connected graph to aggregate the information. The locally connected graph is obtained by a predefined threshold τ in 2 to suppress the redundant edges that have low similarities. In order to show the influence of τ, we conduct several experiments on CGAT with different values of τ, and the classification accuracies are shown in Fig. 10. From the figure, it can be concluded that CGAT obtains the best performance when τ is set to 0.7.

c) Number of Neighborhood Nodes: We select z neighborhood nodes for providing the context information so as to obtain discriminative AFs. In order to determine the influences of different values of z, we conduct several experiments and show their classification accuracies in Fig. 11. From the figure, we can see that CGAT has the best performance when z is equal to 3.

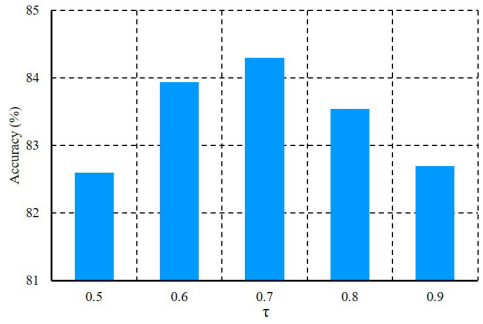


Fig. 10. Classification accuracies of CGAT with different values of τ.

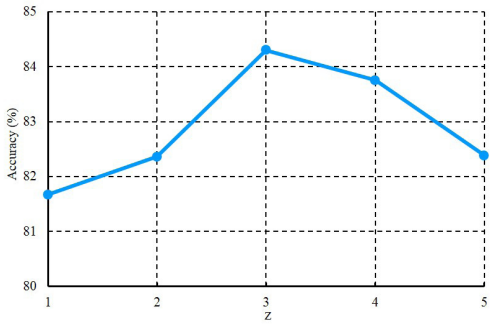


Fig. 11. Classification accuracies of CGAT with a different number of neighborhood nodes.

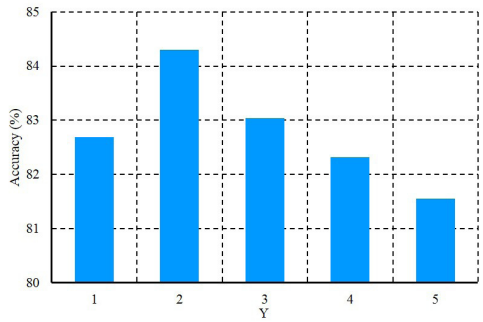


Fig. 12. Classification accuracies of CGAT with a different number of CACs.

d) Number of CAC: In the proposed CGAT, we apply the multihead strategy in the last CGA layer to capture rich information from different aspects. The number of CAC in multihead strategy is Y. We list the classification accuracies of CGAT with different values of Y in Fig. 12, in which the classification accuracy increases until Y increases to 3, and then, the classification accuracy begins to decrease. As a result, three independent attention coefficients are recommended when using the multihead strategy in CGAT.

e) Coefficient μ for the Final Representation: After obtaining the AFs, we concatenate them with the DFs to represent cloud images, and the coefficient μ in 13 is applied to balance the importance of DFs in the final representation. We conduct experiments with different values of μ, and the results are shown in Fig. 13. It shows that when μ is set to 0.9, the best classification accuracy is achieved.

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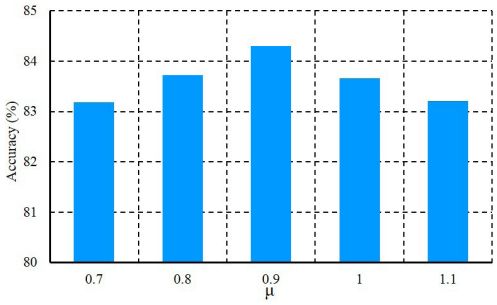


Fig. 13. Classification accuracies of CGAT with different values of μ.

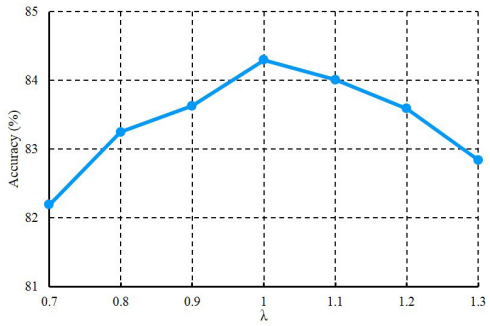


Fig. 14. Classification accuracies of CGAT with different values of λ.

f) Coefficient λ for the Objective Function: The coefficient λ in 15 is utilized to determine the importance of L_{cnn}, and the experimental results with different values of λ are listed in Fig. 14, where CGAT obtains the best performance when λ is equal to 1.

V. Conclusion

In this article, we have proposed the CGAT for ground-based remote sensing cloud classification. Specifically, we first learn the DFs of cloud images using CNN to construct the graph, and then, we propose the CGA layer to obtain the AF of each node. The CAC and two different transformation matrices in the CGA layer are of great importance for obtaining the stable and discriminative AFs of cloud images. In CGAT, we concatenate the AFs with the DFs as the final features that include structure and texture information of cloud images and correlations among cloud images. In addition, we release the largest GCD, which is a necessary precondition to promote the research of ground-based remote sensing cloud classification. The experiments on GCD have shown the effectiveness of CGAT.

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